

are applied to the bases of transistors T9 and T10, respectively. Transistors T11 and T12 amplify these outputs to

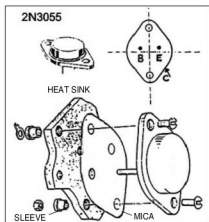


Fig. 2: Mounting of transistor 2N3055 on heat-sink

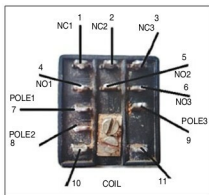


Fig. 3: Details of the 3C/O relay

about 2.2V, which are applied to base terminals of transistors T4 and T8 for further amplification to 12V. CON1 and CON2 are used to connect outputs from the oscillator section to the two transistor banks.

The circuit is powered by a regulated 5V DC provided by voltage regulator 7805. Input to the regulator is the battery voltage, which we get by connecting CON5 to CON6. Battery voltage reaches pin 1 of regulator through pin 9 and pin 3 of relay RL1 and switch S2 is closed. When mains voltage is present, pin 9 and pin 3 are disconnected due to activation of the relay. Power supply to the oscillator section is interrupted, resulting in deactivation of the inverter circuit.

Relay RL1

Relay RL1 affects the necessary changeover required in the system due to the presence or non-presence of mains voltage. It is a 12V, three contacts changeover (three-poles double-throw) relay. The arrangement of poles and contacts is shown in Fig. 3.

The coil of the relay is between terminals 10 and 11. Terminals 7 and 8 are shorted. Connections of remaining terminals of the relay are shown in circuit diagram (Fig. 1).

Working of the circuit

The UPS works in two modes:

1. When AC power is present
2. When AC power is absent

When AC power is present.

When AC mains power is present, transformer X1 gets 230V AC input mains supply. Relay RL1 is therefore energised. Terminals 7, 8 and 9 of the relay come into contact with terminals 4, 5 and 6, respectively. Phase of the incoming AC mains supply gets connected to terminals 4, 7, 5 and 8 of the relay and the output socket where we connect the load. In this manner, mains are transferred to output socket CON8 of the UPS.

When switch S1 is closed, the phase of the input mains gets connected to 230V tapping of transformer X2 through terminal 5 as it is in contact with the terminal 8 of the relay. As the neutral connection is common, transformer X2 acts as a step-down transformer. 230V AC is stepped down to 15V-0-15V AC and rectified to DC voltage by a full-wave rectifier (diodes D3-D4). Capacitor C7 is connected across the center tap of transformer X2. It is not included in the PCB. It is recommended to use a current limiter (say 4.7-ohm, 20W resistor) in series with positive

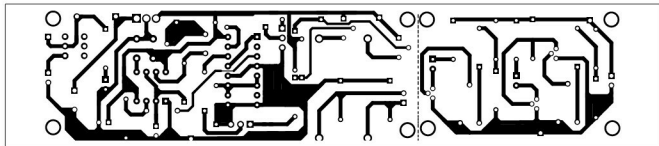


Fig. 4: PCB of the oscillator and the battery-status section

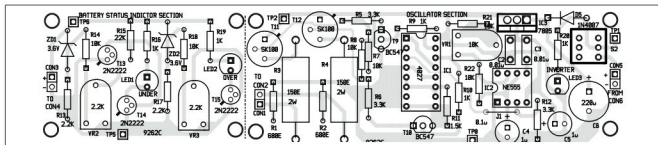


Fig. 5: Components of the PCB